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1. Introduction

Optical coherence tomography angiography (OCTA) is an extension of OCT that visualizes vascular perfusion using the motion contrast from red blood cells (RBCs). Over the years, OCTA has demonstrated widespread acceptance and use for clinical evaluations of vascular features in relation to disease [1]. OCT angiograms can be used to evaluate characteristics on the scale of vascular layers by measuring vascular density or non-perfusion zones, or it can be used to examine individual vessels with measurements of diameter or tortuosity. Nevertheless, OCT is prone to certain artifacts which specifically affect visualization of the vasculature. These artifacts may lead to underestimations of measurements of vessel diameter and density. When viewed in cross-section, large vessels often demonstrate an hourglass appearance with stronger scattering in the center than at the sides (Figure 1A). This appearance is due to orientation effects of RBCs, which align with the blood flow, causing a reduced scattering cross-section at the sides of the vessels. To compensate for these artifacts, OCT contrast agents such as Intralipid have previously been used to increase intravascular intensity signals, particularly in regions where RBC scattering is diminished. Here we examine different methods of measuring vessel size from in vivo OCT scans of mouse retinal vasculature, and we use a contrast agent to obtain ground truth vessel diameter measurements for comparison. We finally propose ways of mitigating or correcting for underestimations of vessel diameter related to typical OCT artifacts.

Optical coherence tomography (OCT) is a non-invasive optical imaging method, which uses interferometry to measure the transit time of light that is backscattered from a sample. OCT has found widespread clinical use in ophthalmology with XXX number of systems and XXX OCT scans performed annually around the world. OCT angiography (OCTA) is a functional extension of OCT which highlights motion between repeated OCT scans to generate images of vascular networks. Typically, OCTA signals are first calculated from the OCT intensity and/or phase information, segmented in 3D manually, algorithmically, or using a deep learning approach, and finally compressed into a 2D image using a maximum intensity projection. These 2D vascular maps can then be binarised by using a global or adaptive threshold, and finally vascular metrics

such as vessel density and diameter can be computed. Of these processing stages, the thresholding step is perhaps the most important. There are dozens of different thresholding methods that have been proposed over the years and each of these may produce different results in the binary map, which would in turn influence the quantification of vascular metrics. In this work, we examine an alternate approach for extracting quantitative vessel diameter by fitting a model to the OCTA data before binarisation to avoid thresholding-based effects.

In addition to thresholding-based variability, we hypothesize that OCTA metrics are also affected by OCT artifacts caused by the non-uniform backscattering profiles of red blood cells (RBCs). Because RBCs have a distinct bi-concave disc shape, they more effectively backscatter light when aligned perpendicular to the beam, rather than parallel to it. It has already been observed that in large vessels, a so-called hourglass artifact appears when viewed cross-sectionally. This is caused by the alignment of the RBCs with the vessel walls while under laminar flow, where the RBCs are perpendicular to the beam in the center of the vessels and parallel to the beam at the edges of the vessels. This artifact artificially suppresses the OCT and OCTA signals at the edges of the vessels, making them appear more narrow when projected into an en-face view, which is the preferred way of displaying angiographic data. Here we use contrast enhanced OCT (CE-OCT) to investigate how significant these effects are and how they can be corrected.

2. Methods and Materials

2.1. Imaging system

All imaging was performed using a custom-built polarisation-sensitive spectral-domain OCT ophthalmoscope, described in detail previously [2]. The system was centred at 840 nm with a spectral bandwidth of 100 nm at full width half maximum (FWHM), providing an axial resolution of approximately 5.1 μm in air and 3.8 μm in tissue (assuming a refractive index of 1.35). A sensitivity of 96 dB was achieved at an A-scan rate of 80 kHz and a beam diameter of 0.5 mm at the pupil plane (power at cornea = 2.85 mW) [3]. This system was employed to acquire volumetric OCT data from mouse retinas before, during, and after an intravascular contrast agent administration, as well as before and after functional stimulation of the mouse retina using a custom 502 nm light-emitting diode (LED) ring flashing at 10 Hz (power at cornea = 26.6 μW). This wavelength was selected to target the peak absorption of mouse rhodopsin (~500 nm), enabling predominantly rod-driven photoreceptor stimulation [4].

2.2. Experimental protocol

Three mouse models were used in this study: VLDLR knockout mice (B6;129S7-Vldlr^{tm1Her}/J, The Jackson Laboratory, Bar Harbor, USA), C57BL/6 mice (C57BL/6, The Jackson Laboratory, Bar Harbor, USA), and SOD1 transgenic mice (strain details, The Jackson Laboratory, Bar Harbor, USA), with $n = \text{XX}$ mice per group. All imaging sessions were conducted under general anaesthesia using either isoflurane or a ketamine/xylazine combination. Isoflurane was induced at 4% in oxygen for 4 minutes and subsequently maintained at 2% in oxygen at a flow rate of 1.5–2 L/min. In cases where isoflurane alone did not provide adequate sedation, an intraperitoneal injection of ketamine/xylazine cocktail (100 mg/kg ketamine, 6 mg/kg xylazine; 10 mL/kg body weight) was administered. To maintain core body temperature of the mice throughout imaging, they were positioned on a heating pad. Mouse pupils were dilated with one drop of tropicamide (5 mg/mL, Agepha Pharma s.r.o., Senec, Slovakia) per eye, and artificial tear drops (Oculotect, ALCON Pharma GmbH, Freiburg im Breisgau, Germany) were applied at regular intervals to prevent the cornea from drying out.

Intralipid has been shown to be a highly scattering contrast agent that fills the vascular lumen uniformly, thereby compensating for the artefact caused by the diminished backscattering at vessel peripheries caused by RBC orientation effects [3,5,6]. For this study, Intralipid 20% (3 mL/kg

body weight) was administered as a bolus injection via the tail vein. Volumetric angiography data were acquired over a 1 mm × 1 mm field of view centred on the optic nerve head (ONH), with a scan density of 500 A-scans per B-scan and five repeated B-scans at each of 400 slow-axis positions (acquisition time approximately 16 seconds). For this study, pre- and post-injection datasets and pre- and post-stimulation datasets were acquired. **These volumetric data were used solely for qualitative visualisation of the contrast enhancement effect across the retinal vascular plexuses and were not used for quantitative diameter analysis.** Additionally, a dynamic-contrast OCT (DyC-OCT) protocol [7] was employed during the contrast injection or during stimulation, consisting of 2000 repeated B-scans over approximately 16 seconds (interscan time 8.2 ms) at the same cross-sectional location over a 1 mm lateral field of view. DyC-OCT scans allow us to track the same vessels over time to precisely measure the effects of the contrast agent or the stimulation with a high degree of spatial coregistration. **All quantitative vessel diameter measurements reported in this study were derived from the DyC-OCT data.**

Functional flicker stimulation was performed in a C57BL/6 mouse to assess whether the choice of diameter measurement method affects the detection of physiological vascular responses. For this experiment, ketamine/xylazine cocktail was chosen over isoflurane to minimise anaesthesia-induced vasodilation. A DyC-OCT acquisition was performed during which the retina was unstimulated for approximately 4 seconds (baseline), followed by 10 Hz flicker stimulation from the LED ring for approximately 7.5 seconds, and a post-stimulation period of approximately 4.5 seconds **(I wrote this to match the pre- and post-stim lines in the stimulation result plots) CM[This isn't the stimulation protocol that was used though. The lines just mark where we were averaging, but I believe in this data set there was already stimulation before the data started acquiring and it continued through the full acquisition. Do you remember how much this was?].**

All animal procedures were approved by the ethics committee of the Medical University of Vienna and the Austrian Federal Ministry of Education, Science, and Research (BMBWF/66.009/0272-V/3b/2019 and GZ 2024-0.802.524).

2.3. Post-processing and analysis

Cross-polarised OCT intensity data from the retinal pigment epithelium (RPE) were used for inter-frame alignment and B-scan flattening [3, 8]. Angiograms were derived by applying the complex subtraction OCT angiography (OCTA) algorithm [9, 10]. We evaluated multiple combinations of methods for quantifying vessel diameter from the OCTA data to determine which combinations of steps work best under different conditions. Specifically, we looked at two types of data projections (maximum intensity and mean intensity) to generate vessel profiles and several different quantifications of those profiles including a traditional binarisation, a Gaussian model approach, and a Generalised-Gaussian (GG) model approach. These methods represent commonly used methods in the literature for OCTA quantification [11–14]. The different methods were evaluated against the ground truth as described in this section. All data processing and analysis was performed in MATLAB (R2019b, MathWorks, Natick, MA, USA).

2.3.1. Ground truth calculation

Ground truth vessel diameters were obtained from contrast-enhanced DyC-OCT B-scan angiograms, where the uniformly scattering Intralipid fills the full vascular lumen regardless of RBC orientation. The width and the height of the vessel were manually measured from the vessel's cross-section. The ground truth diameter was defined as the minimum of the two measurements to account for off-axis cross-sectioning of the vessel, which produces a more elliptical appearance in the B-scan. For cases where the vessel is perpendicular to the B-scan, the height and width would therefore be in agreement. Because the lateral diameter is studied here, we defined a correction factor of *height/width* for cases where the width was larger than the height due to skew effects and was equal to 1 otherwise. By multiplying this correction factor against diameter

141 measured from the ROI projection, we ensured that the measured vessel diameter reflected the
 142 true size of the vessel.

143 2.3.2. Vessel projection calculation

144 For each vessel of interest, a rectangular region of interest (ROI) encompassing the vessel cross-
 145 section, along the lateral (x) and axial (z) dimensions, was selected from the DyC-OCT B-scan
 146 angiograms. En face intensity profiles were then generated by projecting the A-scans within this
 147 ROI along the axial (z) direction yielding a time-stack of 1-D lateral intensity profiles for each
 148 vessel. Two projection methods were compared: maximum intensity projection (MIP), which
 149 selects the highest intensity value along the depth axis of each ROI, and mean intensity projection
 150 (MeIP), which averages the values along each ROI. Each projection resulted in a time-stack of
 151 one-dimensional intensity profiles for each vessel, which were then used for subsequent diameter
 152 quantification. For the model-based approaches, the profiles were normalised to a range of 0–1
 153 to ensure a good fit.

154 2.3.3. Model-based vessel diameter calculation

155 The two parametric models were fitted (Fig. 1a) to the one-dimensional intensity profiles from
 156 the previous step, using nonlinear least-squares regression in MATLAB.

157 Standard Gaussian function:

$$f(x) = A \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) + C \quad (1)$$

158 where A is the peak amplitude, μ is the centre position, σ is the standard deviation, and C is a
 159 constant baseline offset.

160 Generalised-Gaussian (GG) function:

$$f(x) = A \exp\left(-\left(\frac{|x - \mu|}{\alpha}\right)^\beta\right) + C \quad (2)$$

161 where A is the peak amplitude, μ is the centre position, α is the scale parameter, β is the shape
 162 parameter, and C is a constant baseline offset. Given the imaging system configuration, all vessels
 163 are expected to produce at minimum a Gaussian intensity profile in the B-scan cross-section,
 164 and so profiles sharper than Gaussian profile ($\beta < 2$) are therefore not technically meaningful.
 165 The shape parameter β was hence constrained to $\beta \geq 2$ during fitting. With this constraint, the
 166 GG accommodated to profiles ranging from Gaussian ($\beta = 2$, where $\sigma = \alpha/\sqrt{2}$) to increasingly
 167 flat-topped ($\beta > 2$).

168 Two standard width metrics were calculated from the fitted models: the full width at half
 169 maximum (FWHM) and the $1/e^2$ width. For the standard Gaussian fit, the $1/e^2$ width was
 170 calculated as:

$$w_{1/e^2, G} = 4\sigma \quad (3)$$

171 The FWHM for a Gaussian is related to the $1/e^2$ width by a fixed proportionality:

$$\text{FWHM}_G = 2\sigma\sqrt{2\ln 2} = \frac{\sqrt{2\ln 2}}{2} w_{1/e^2, G} \quad (4)$$

172 and therefore only the $1/e^2$ width was calculated directly from the fit, with the FWHM obtainable
 173 via this known relationship.

174 For the GG fit, no such linear relationship exists between the FWHM and $1/e^2$ width, since
 175 both depend on the shape parameter β . Both metrics were therefore calculated independently as:

$$\text{FWHM}_{GG} = 2\alpha(\ln 2)^{1/\beta} \quad (5)$$

176

$$w_{1/e^2, GG} = 2\alpha \cdot 2^{1/\beta} \quad (6)$$

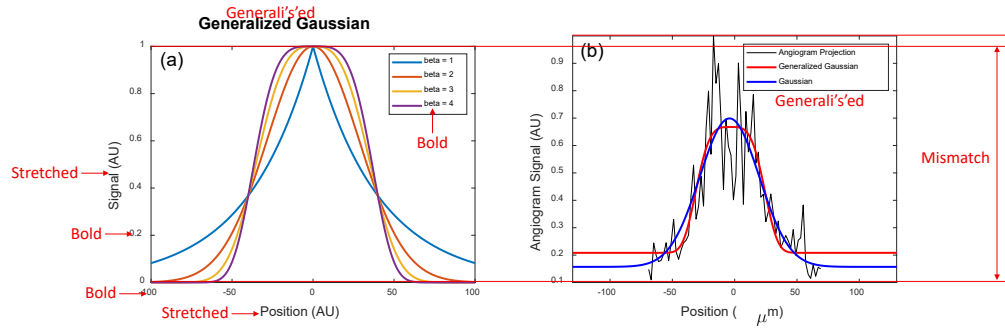


Fig. 1. (a) GG function for shape parameters $\beta = 1, 2, 3$, and 4 . At $\beta = 2$ the function reduces to a standard Gaussian; higher β values produce increasingly flat-topped profiles. (b) Example en face vessel intensity profile (black) with standard Gaussian (blue) and GG (red) fits.

2.3.4. Binary mask based vessel diameter calculation

For comparison with OCTA processing pipelines that rely on image binarisation, vessel diameters were also measured from binary masks generated by global thresholding. A 3D averaging filter with a kernel size of $50 \times 50 \times 10$ voxels (lateral $x \times$ axial $z \times$ temporal t) was applied to the DyC-OCT angiogram volumes to reduce noise and minimize the effects of outliers. To produce a binarised black and white (BW) mask, the maximum angiogram intensity within these smoothed volumes were calculated and five different threshold values were defined as fixed percentages of these maxima: **BW-1 = XX%, BW-2 = XX%, BW-3 = XX%, BW-4 = XX%, and BW-5 = XX%**. Each threshold was applied to the unsmoothed en face projection (MIP and MeIP) to generate a binary mask, and the vessel diameter was measured as the number of pixels above the threshold across the lateral vessel cross-section.

2.3.5. Statistical analysis

To evaluate the accuracy and precision of all diameter measurement approaches, the calculated diameter values were plotted against the ground truth for each vessel across three conditions: pre-contrast, peak contrast, and post-contrast. A linear regression was fitted to each condition, yielding the slope (m) and coefficient of determination (R^2). A slope of $m = 1$ indicated perfect agreement with the ground truth, and R^2 quantified the goodness of fit of the linear model. Additionally, the coefficient of variation (CoV) was computed for each method to assess measurement precision (repeatability). For the model-based methods, six combinations were evaluated: two projection methods (MIP and MeIP) by three width metrics ($w_{1/e^2, G}$, $FWHM_{GG}$, and $w_{1/e^2, GG}$). For the binarisation-based methods, diameter was measured at five threshold levels (BW-1 through BW-5) for both MIP and MeIP. For the functional flicker stimulation experiment, the Pearson correlation coefficient (ρ) between measured diameter values and angiogram intensity was computed for each method and vessel, to assess whether the diameter measurements were affected by changes in angiogram intensity. Measurements with $p < 0.05$ were considered statistically significant.

3. Results

3.1. Qualitative assessment of vessel underestimation

Figure 2 shows OCTA MIP enface images of the superficial vascular plexus (SVP), intermediate capillary plexus (ICP), and deep capillary plexus (DCP) before and after contrast injection. Across

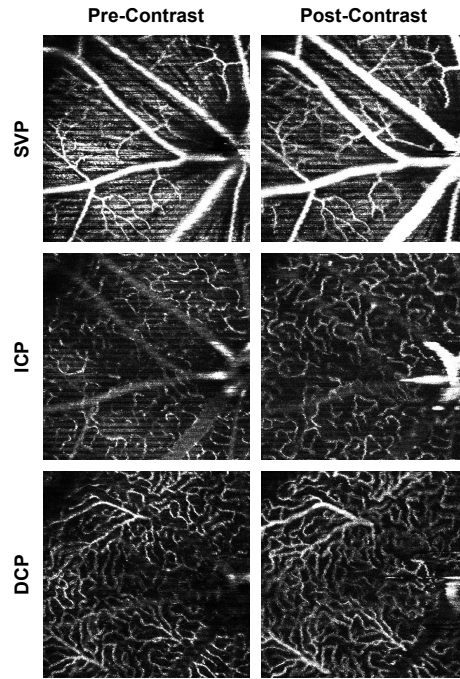


Fig. 2. OCTA MIP en face of the SVP, ICP, and DCP before (left) and after (right) the administration of Intralipid 20%. Vessels appear visibly wider and brighter post-contrast across all layers, most prominently in the larger vessels.

all three layers, vessels appeared visibly wider and brighter post-contrast injection, consistent with previous literature showing that OCTA underestimates vessel diameter in the absence of a contrast agent [3]. This effect was most pronounced in the larger vessels, especially in the SVP, where the major retinal arteries and veins appeared noticeably thicker after contrast administration. In the ICP and DCP, the capillary networks also showed improved visibility, though the effect was less dramatic.

DyC OCTA projections helped visualise the temporal changes in vessel diameter pre- and post-contrast injection (Fig. 3). Pre-contrast injection, the vessel cross-sections appeared narrower than their true diameter (Fig. 3a), which could be due to lateral edge signal suppression by the RBC orientation artefact [15]. At peak contrast, the Intralipid filled the vascular lumen uniformly and showed the full extent of the vessel cross-section (Fig. 3b). However, the high scattering nature of the contrast agent produced light diffusion beyond the vessel wall, resulting in an apparent vessel diameter that slightly exceeded the true anatomical diameter. In the post-contrast phase, the signal stabilised as the initial bolus cleared, and the vessel boundaries remained more visible than before injection without the overestimation observed at peak contrast (Fig. 3c). The MIP and the corresponding binary mask over time (Figs. 3d–e) showed the observed temporal changes in vessel diameter across the three phases. The mean OCTA intensity (Fig. 3f) showed a sharp increase post-contrast injection, followed by a decrease to a new steady state that remained elevated above the pre-contrast baseline. These observations confirm that the contrast agent does significantly enhance vessel visibility and diameter, and that the peak- and post-contrast images show improved vessel delineation compared to pre-contrast.

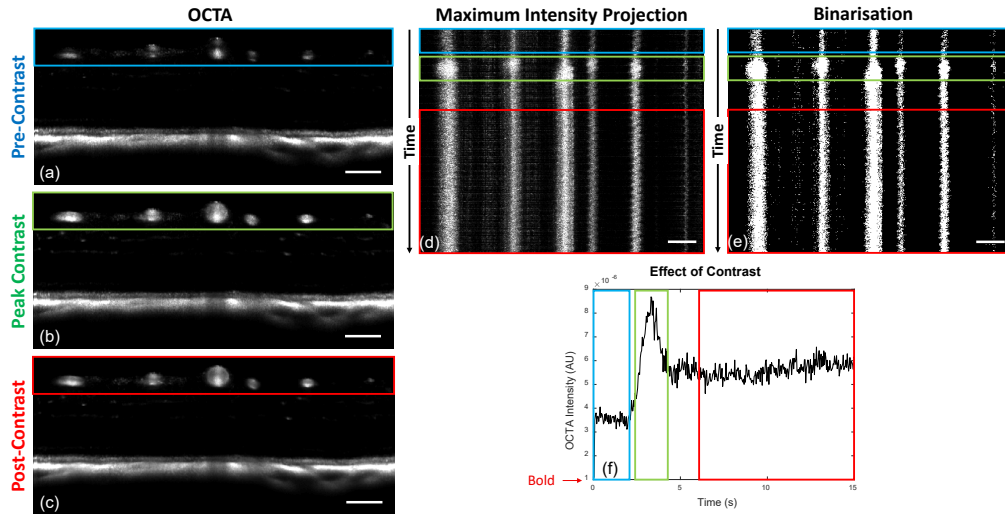


Fig. 3. DyC-OCT B-scan angiograms at (a) pre-contrast, (b) peak contrast, and (c) post-contrast timepoints. (d) MIP and (e) binarisation of the vessel cross-sections over time, with coloured boxes indicating the pre-contrast (blue), peak-contrast (green), and post-contrast (red) windows. (f) Mean OCTA intensity over time, showing the temporal intensity dynamics. All scale bars are 100 μm .

3.2. Accuracy and precision of diameter measurements

The accuracy (m , R^2) and precision (CoV) of all diameter measurement methods are summarised in Table 1.

All model-based methods underestimated vessel diameter in pre-contrast data, with slopes ranging from 0.42 to 0.73 (Table 1). Contrast enhancement improved accuracy across all metrics, though the degree of improvement varied. The FWHM_{GG} metric remained well below unity even post-contrast. The $w_{1/e^2, \text{G}}$ with MIP slightly overestimated post-contrast ($m = 1.12\text{--}1.15$), while MeIP yielded values closer to unity ($m = 1.04$). The $w_{1/e^2, \text{GG}}$ with MeIP provided the best overall agreement, achieving a slope closest to unity with $m = 1.001$ ($R^2 = 0.97$) post-contrast, which was the closest to the identity line of all tested model-based methods. Across all metrics, MeIP mostly outperformed MIP in both accuracy and linearity (Table 1, Fig. 4).

For binary mask based methods, the slope decreased with increasing threshold across all contrast conditions and projections (Table 1). This degradation was steeper for MIP than for MeIP. MIP slopes dropped from near unity at BW-1 to well below it at BW-5, whereas MeIP slopes declined more gradually and remained closer to unity across the threshold range. MeIP slopes were consistently higher than MIP at every threshold and contrast condition, though at lower thresholds and at peak and post-contrast, MeIP tended to slightly overestimate vessel diameter. The R^2 values followed a similar pattern. For MIP, pre- and post-contrast R^2 decreased substantially with increasing threshold, whereas for MeIP, R^2 remained relatively stable. Overall, MeIP provided more reliable binarisation-based diameter measurements than MIP.

The precision of each method was assessed via the CoV (Table 1, Fig. 7). For MIP, the binary mask based CoV increased substantially with threshold, whereas the model-based CoV remained stable across all contrast conditions. For MeIP, the binary mask based CoV was notably more stable across thresholds than for MIP, and the model-based CoV values were lower overall, though they increased slightly at peak contrast before decreasing post-contrast. Across both methods, MeIP yielded lower CoV values than MIP pre-contrast and more stable values post-contrast. Among the model-based metrics, FWHM_{GG} consistently exhibited the weakest accuracy while

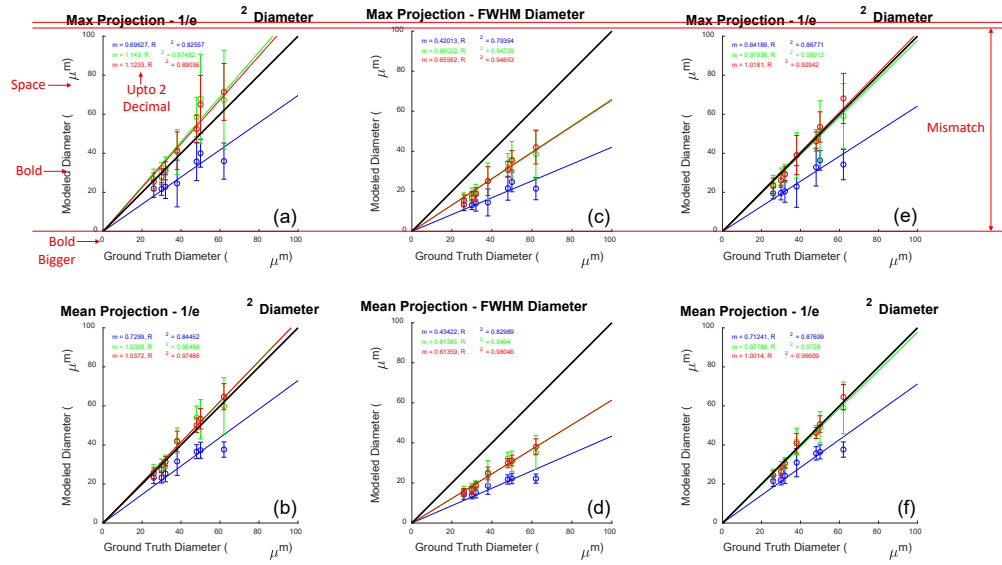


Fig. 4. Modelled versus ground truth vessel diameter using MIP (top row) and MeIP (bottom row). Columns correspond to (a, b) $w_{1/e^2, G}$, (c, d) FWHM_{GG} , and (e, f) $w_{1/e^2, GG}$. Linear regression lines and statistics are shown for pre-contrast (blue), peak contrast (green), and post-contrast (red) conditions. The black line indicates the identity line ($m = 1$).

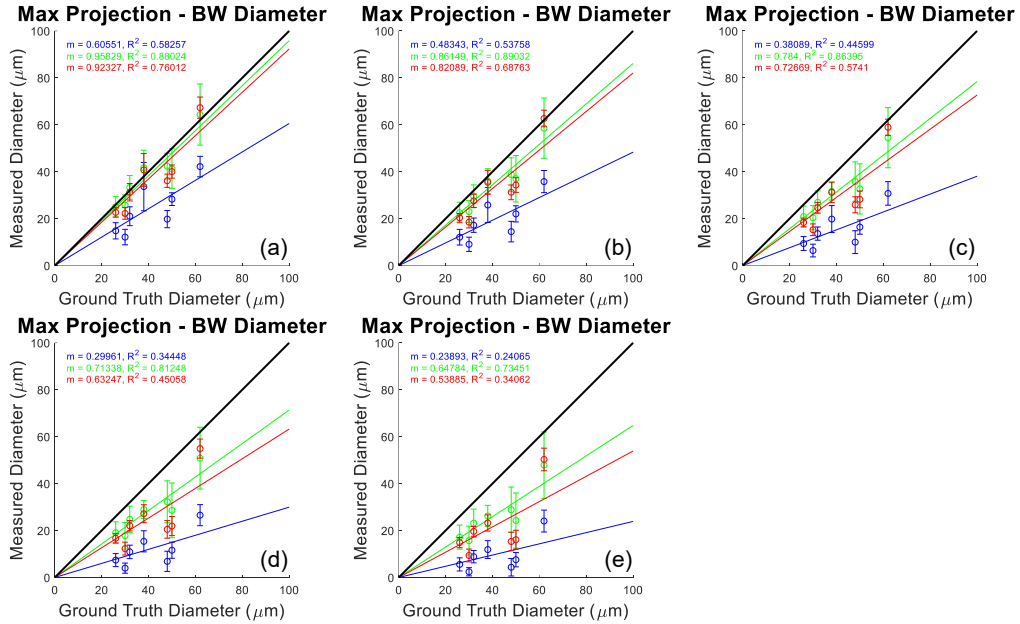


Fig. 5. Vessel diameter measured from binarised masks versus ground truth for MIP at five threshold levels: (a) BW-1 through (e) BW-5. Linear regression lines and statistics are shown for pre-contrast (blue), peak contrast (green), and post-contrast (red) conditions.

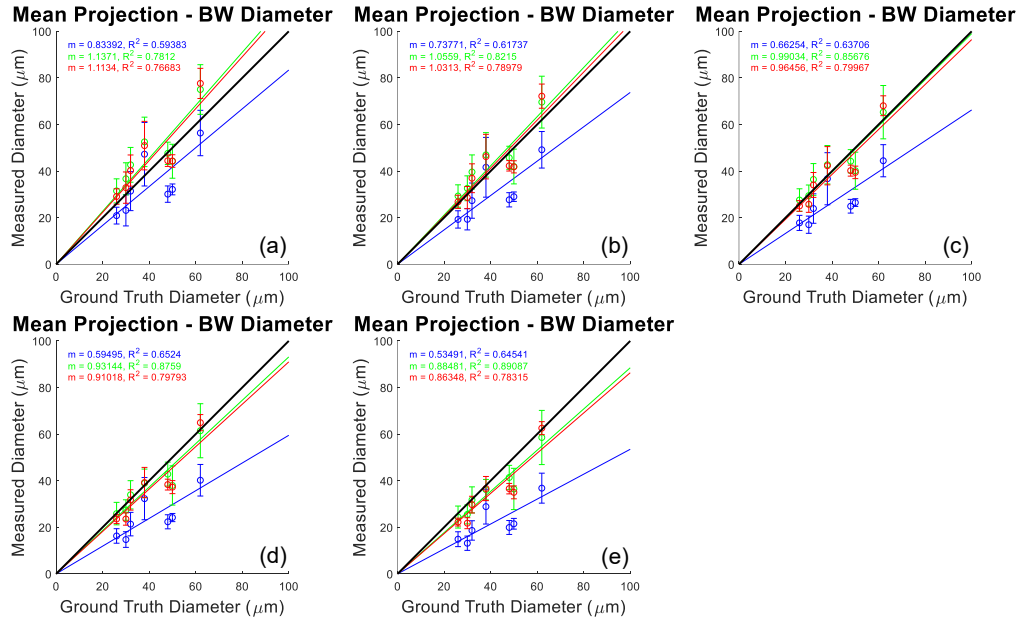


Fig. 6. Vessel diameter measured from binarised masks versus ground truth for MeIP at five threshold levels: (a) BW-1 through (e) BW-5. Linear regression lines and statistics are shown for pre-contrast (blue), peak contrast (green), and post-contrast (red) conditions.

maintaining comparable precision. Overall, model-based methods consistently outperformed binarisation in both accuracy and precision, and MeIP yielded superior performance compared to MIP across all methods.

3.3. Functional flicker stimulation

The model-based diameter time series (Figs. 8a,b) showed increase in diameter post-stimulation for most of the vessels. The binary mask based diameter time series (BW-3; Fig. 8c) showed a mixed trend, where most vessels showed a decrease in diameter post-stimulation. Bar plots of the mean diameter pre- and post-stimulation (Figs. 8d-f) confirmed this pattern. For the Gaussian model, four of five vessels showed significant diameter increases ($p < 0.01$), and the GG model showed significant increases in three of five vessels (Table 2). In contrast, the binary mask based diameter showed significant decreases in four of five vessels.

The angiogram intensity time series (Fig. 8g) showed a visible decline in angiogram intensity during the duration of the DyC-OCT scan across all five vessels. The mean intensity comparison pre- and post-stimulation (Fig. 8h) confirmed that the decrease was significant for all vessels, ranging from -14.1% to -19.9% (Table 2). The binary mask based method showed uniformly positive per-vessel correlations, with an overall correlation of $\rho = 0.523$ ($p < 0.0005$). The Gaussian and GG models yielded overall correlations of $\rho = -0.101$ and $\rho = -0.098$, respectively. These results demonstrated that the model-based diameter calculation provided more accurate and repeatable measurements, and also recovered physiologically meaningful vascular responses that binary mask based methods misrepresented.

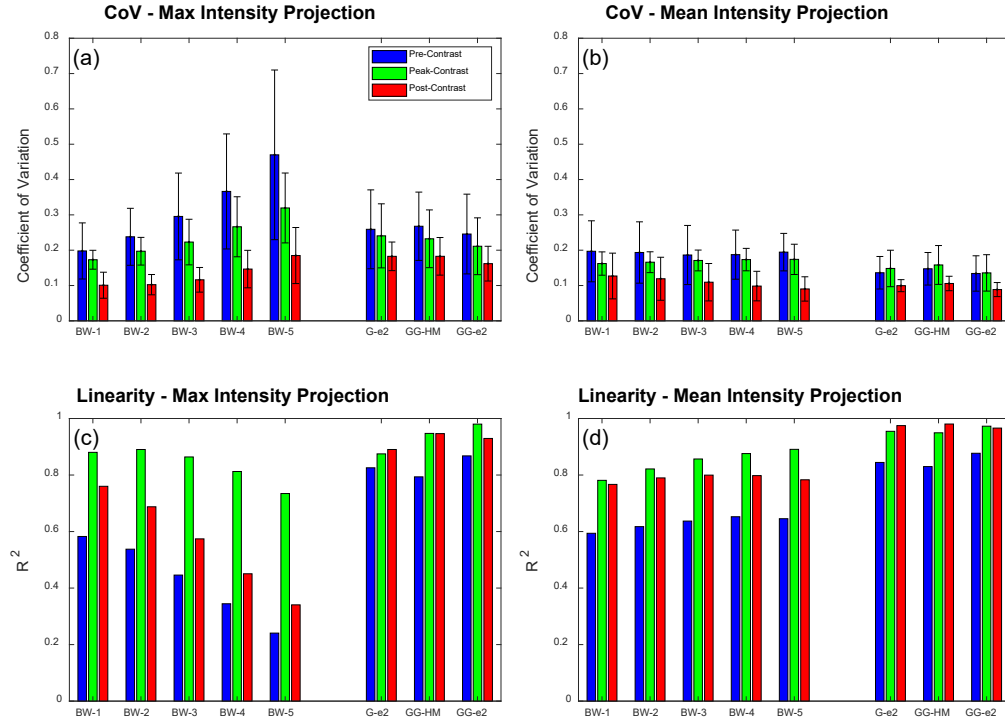


Fig. 7. Summary of accuracy and precision across all diameter measurement methods. (a, b) Coefficient of variation (CoV) and (c, d) coefficient of determination (R^2) for MIP (a, c) and MeIP (b, d), respectively. Bars are grouped by method (BW-1 through BW-5, G-e², GG-HM, GG-e²) and colour-coded by contrast condition: pre-contrast (blue), peak-contrast (green), and post-contrast (red).

Table 1. Summary of slope (m), coefficient of determination (R^2), and coefficient of variation (CoV) for all diameter measurement methods across pre-contrast, peak contrast, and post-contrast conditions. Methods are grouped as binary mask based and model-based. Bold values indicate slope agreement within 10% of unity ($0.9 \leq m \leq 1.1$), strong linearity ($R^2 \geq 0.90$), or high precision (CoV ≤ 0.10). (confirm values with MATLAB.)

	BW-1		BW-2		BW-3		BW-4		BW-5		$w_{1/e^2, G}$		FWHM _{GG}		$w_{1/e^2, GG}$	
	MIP	MeIP	MIP	MeIP	MIP	MeIP	MIP	MeIP	MIP	MeIP	MIP	MeIP	MIP	MeIP	MIP	MeIP
<i>Slope (m)</i>																
Pre	0.61	0.83	0.48	0.74	0.38	0.66	0.30	0.59	0.24	0.53	0.70	0.73	0.42	0.43	0.64	0.71
Peak	0.96	1.14	0.86	1.06	0.78	0.99	0.71	0.93	0.65	0.88	1.15	1.04	0.66	0.61	0.98	0.98
Post	0.92	1.11	0.82	1.03	0.73	0.97	0.63	0.91	0.54	0.86	1.12	1.04	0.66	0.61	1.02	1.00
<i>Coefficient of determination (R^2)</i>																
Pre	0.58	0.59	0.54	0.62	0.45	0.64	0.34	0.65	0.24	0.65	0.83	0.84	0.79	0.83	0.87	0.88
Peak	0.88	0.78	0.89	0.82	0.86	0.86	0.81	0.88	0.74	0.89	0.87	0.95	0.95	0.95	0.98	0.97
Post	0.76	0.77	0.69	0.79	0.57	0.80	0.45	0.80	0.34	0.78	0.89	0.97	0.95	0.98	0.93	0.97
<i>Coefficient of variation (CoV) (values extracted from bar plots, confirm with MATLAB)</i>																
Pre	0.20	0.20	0.24	0.20	0.29	0.19	0.37	0.19	0.47	0.20	0.26	0.14	0.26	0.15	0.25	0.14
Peak	0.17	0.16	0.20	0.15	0.23	0.17	0.27	0.18	0.32	0.18	0.22	0.16	0.23	0.17	0.21	0.15
Post	0.10	0.12	0.10	0.12	0.11	0.11	0.15	0.10	0.19	0.09	0.19	0.10	0.19	0.11	0.16	0.09

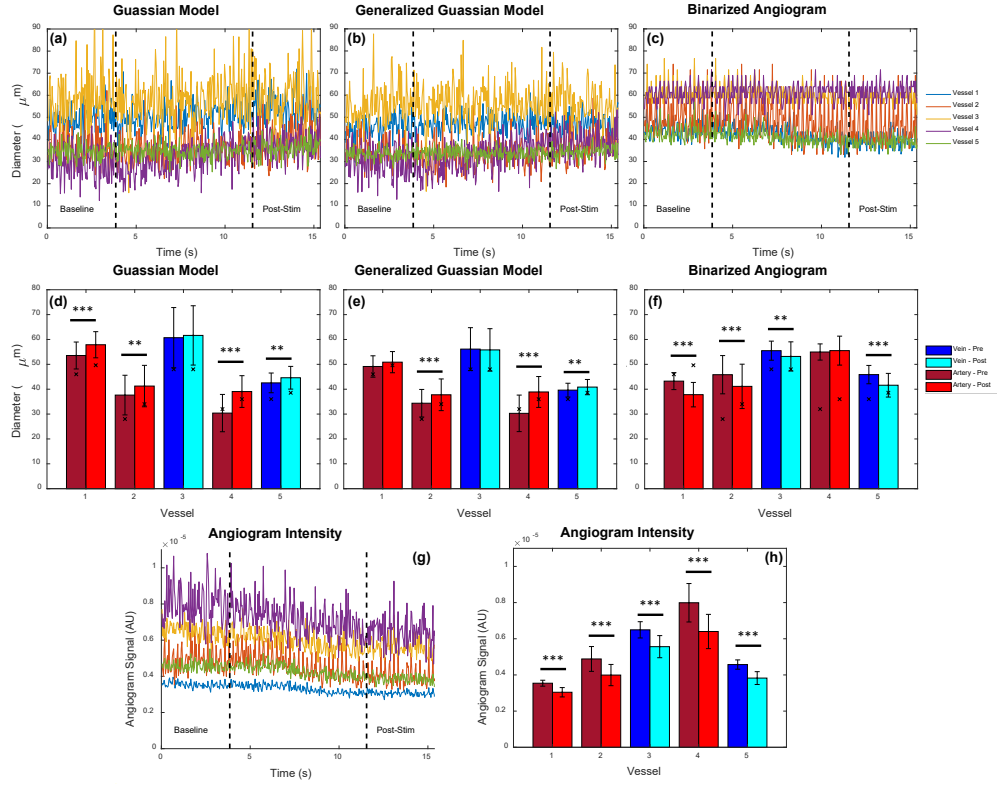


Fig. 8. Functional flicker stimulation results. (a–c) Vessel diameter time series pre- and post-stimulation for (a) Gaussian model, (b) Generalised Gaussian model, and (c) binarised angiogram, across five vessels. Vertical dashed lines indicate stimulation onset and offset. (d–f) Mean diameter per vessel before (dark bars) and after (light bars) stimulation for each method. Veins are shown in blue/cyan and arteries in dark/light red. Significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. (g) Angiogram intensity time series and (h) mean angiogram intensity per vessel pre- and post-stimulation.

Table 2. Flicker stimulation results. Δd : diameter change (%); ρ : Correlation between diameter and angiogram intensity; ΔI : intensity change (%). Statistically significant values are bold (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

	Gaussian		GG		Binarised		Intensity
	Δd (%)	ρ	Δd (%)	ρ	Δd (%)	ρ	
Vessel 1	+9.2***	-0.07	+1.5	0.03	-12.5***	0.59***	-14.1***
Vessel 2	+9.6**	0.23*	+9.9***	0.07	-10.2***	0.80***	-18.2***
Vessel 3	+1.5	-0.28**	-0.6	-0.28**	-4.2**	0.27**	-14.3***
Vessel 4	+28.5***	-0.15	+28.4***	-0.14	+1.0	0.46***	-19.9***
Vessel 5	+4.9***	-0.24*	+3.1**	-0.18	-9.3***	0.52***	-16.3***
Overall ρ		-0.101		-0.098		0.523	

275 4. Discussion

276 4.1. Model-based vs Binary Mask Based Diameter Measurement

277 The model-based and binary mask based diameter approaches define vessel boundary in
278 fundamentally different ways. Binarisation of the en face maps discards the spatial intensity
279 distribution and reduces each pixel to a binary decision. The measured diameter directly depends
280 on the chosen threshold value and on angiogram intensity, both of which may vary across imaging
281 sessions, contrast conditions, and even between vessels within the same scan. Model-based
282 methods fit a parametric model function to the maximum or mean intensity profiles, and calculate
283 the diameter from the fitted parameters. As the fit captures the shape of the profile rather than
284 the absolute intensity, it is inherently less sensitive to changes in angiogram signal and does not
285 require a varying threshold.

286 This difference explains the trends observed in Fig. 7. Model-based methods maintained
287 strong linearity with the ground truth across all contrast conditions, whereas the binary mask
288 based slopes degraded progressively with increasing threshold. Pre-contrast, the RBC orientation
289 artefact suppresses edge signals, causing all methods to underestimate diameter [15]. However,
290 model-based approaches still maintained a linear relationship with the ground truth, producing
291 systematic, predictable underestimation. Binary mask methods on the other hand showed
292 much weaker linearity because the suppressed edge signals fell below the binarisation threshold
293 inconsistently, depending on vessel size and local intensity.

294 The generalised Gaussian (GG) model outperformed the standard Gaussian because the
295 intensity profiles of larger vessels are not purely Gaussian (Fig. ??b). The standard Gaussian
296 cannot capture relatively flat-topped angiogram intensity profiles and tends to fit a narrower
297 curve, whereas the GG model accommodates it through the shape parameter β , which ranges
298 from Gaussian ($\beta = 2$) to increasingly flat-topped profiles ($\beta > 2$). Among the width metrics,
299 FWHM_{GG} consistently showed the weakest accuracy. It could be because the half-maximum
300 places the vessel boundary where the intensity has already dropped inside the true edge. The
301 $1/e^2$ width might capture more of the transition region and therefore may have provided a more
302 representative measure of the vessel diameter.

303 4.2. Best Choices Under Different Conditions

304 When a contrast agent is available and accuracy is the primary goal, the $w_{1/e^2, \text{GG}}$ metric with
305 MeIP is the clear recommendation. Post-contrast, this combination achieved a slope of $m = 1.00$
306 with $R^2 = 0.97$ and a CoV of 0.09, representing the best accuracy, linearity, and precision of any
307 tested method. When only pre-contrast data are available, the same metric still provides the best
308 linearity ($R^2 = 0.88$) among all methods, though the slope ($m = 0.71$) indicates that diameter
309 will be underestimated by approximately 30%. In this case, a correction factor could be applied if
310 absolute diameter values are needed, though relative comparisons between vessels or timepoints
311 would remain valid without correction.

312 When only maximum intensity projection data are available, as is the case for many commercial
313 OCTA instruments and existing datasets, the $w_{1/e^2, \text{GG}}$ with MIP still provides good post-contrast
314 accuracy ($m = 1.02$, $R^2 = 0.93$). For binary mask based analysis of MIP data, only the lowest
315 threshold (BW-1) at post-contrast achieved a slope near unity ($m = 0.92$), and the slope degraded
316 rapidly with increasing threshold. This highlights that if binarisation must be used with MIP data,
317 a very low threshold is essential, though this comes at the cost of increased noise sensitivity.

318 When computation time is a limiting factor and model fitting is not practical, MeIP with a
319 moderate threshold (e.g. BW-3) provides a reasonable alternative. At post-contrast, BW-3 with
320 MeIP achieved $m = 0.97$ and $R^2 = 0.80$, which is acceptable for many applications. Additionally,
321 MeIP-based binarisation showed more stable CoV values across thresholds compared to MIP,
322 making the results less sensitive to the exact threshold choice. This stability makes MeIP the

323 preferred projection for any binarisation-based workflow.

324 4.3. *Maximum Intensity or Mean Intensity Projection?*

325 Across nearly all methods and conditions, MeIP outperformed MIP in both accuracy and precision
326 (Table 1). This finding is noteworthy because MIP has been shown to produce higher signal-
327 to-noise ratio (SNR) and contrast in en face OCTA images [12], and is the default projection
328 method in many commercial OCTA systems and analysis pipelines [11, 13]. However, higher
329 SNR does not necessarily translate to more accurate diameter measurements.

330 The MIP selects the single highest intensity value along the depth axis of each A-scan within
331 the vessel ROI. This makes it inherently sensitive to outliers like a single bright voxel, whether
332 from a peak-intensity RBC event, noise, or a transient scattering enhancement, will dominate the
333 projected profile. MeIP, by contrast, averages all intensity values along the depth axis, which
334 smooths out frame-to-frame intensity fluctuations and produces a more representative profile
335 of the average vessel cross-section. This averaging effect is particularly beneficial for diameter
336 measurement, where the goal is to capture the typical vessel boundary rather than the most
337 extreme signal event.

338 The difference between MIP and MeIP was especially apparent in the binary mask based
339 results. For MIP, the slope decreased steeply with increasing threshold, dropping from 0.92 at
340 BW-1 to 0.54 at BW-5 post-contrast. For MeIP, the slope declined more gradually and remained
341 above 0.86 even at BW-5 post-contrast. This indicates that MeIP produces a more uniform
342 intensity distribution across the vessel profile, making the binarisation result less sensitive to the
343 exact threshold value. The CoV data reinforced this: MIP-based binary mask CoV increased
344 substantially with threshold, while MeIP-based CoV remained relatively stable.

345 For model-based methods, MeIP also showed improved performance. The $w_{1/e^2, G}$ with
346 MIP slightly overestimated vessel diameter post-contrast ($m = 1.12$), likely because the MIP
347 picks up the brightest frame where the signal may extend slightly beyond the vessel boundary.
348 MeIP reduced this to $m = 1.04$, a much closer agreement with the ground truth. These results
349 suggest that while MIP may be preferred for visualisation purposes where contrast and SNR
350 are important [12], MeIP should be the preferred projection method when quantitative diameter
351 measurements are the goal.

352 4.4. *Influence of Intensity on Measurements*

353 The functional flicker stimulation experiment provided a direct demonstration of how changes in
354 angiogram signal intensity can affect diameter measurements differently depending on the method
355 used. During the DyC-OCT acquisition, the angiogram intensity declined significantly across all
356 five vessels (Fig. 8g,h), with decreases ranging from -14.1% to -19.9% (Table 2). This decline
357 is expected during prolonged imaging sessions due to factors such as drying of the cornea and
358 subtle changes in alignment. Simultaneously, the physiological response to flicker stimulation
359 should produce vasodilation, as has been well established in retinal physiology [16, 17].

360 The model-based methods correctly detected this expected vasodilation, with the Gaussian
361 model showing significant diameter increases in four of five vessels (up to $+28.5\%$) and the GG
362 model showing significant increases in three of five vessels (Table 2). The binary mask based
363 method, however, reported significant diameter decreases in four of five vessels, with changes
364 ranging from -4.2% to -12.5% . This is a paradoxical result: the binary mask based method
365 reported vasoconstriction in a stimulation paradigm that is known to produce vasodilation.

366 The explanation lies in how each method responds to declining signal intensity. For the binary
367 mask based method, the threshold is fixed and diameter is determined by the number of pixels
368 that exceed this threshold. When the overall angiogram intensity drops, fewer pixels at the
369 vessel edges exceed the threshold, and the vessel appears to shrink — even if the physical vessel
370 diameter has increased. This is confirmed by the strong positive correlation between measured

371 diameter and angiogram intensity for the binary mask based method ($\rho = 0.523$, $p < 0.0005$). In
 372 contrast, the model-based methods showed weak overall correlations with intensity ($\rho = -0.101$
 373 for Gaussian, $\rho = -0.098$ for GG), indicating that the fitted diameter was effectively independent
 374 of the signal intensity (Fig. 9).

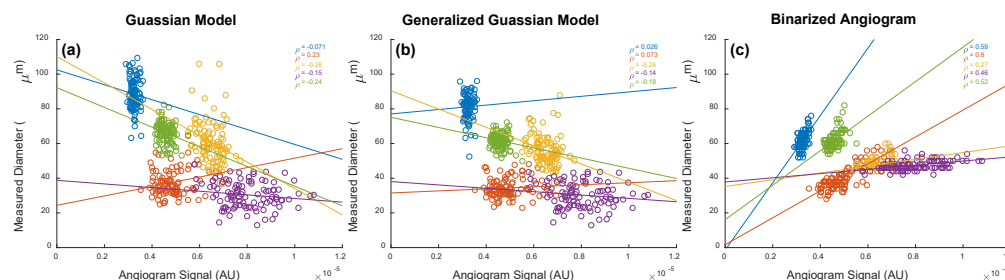


Fig. 9. Measured vessel diameter as a function of angiogram signal intensity during functional flicker stimulation for (a) Gaussian model, (b) generalised Gaussian model, and (c) binarised angiogram. Each colour represents one of five vessels; ρ denotes the Pearson correlation coefficient. The binarised method shows strong positive correlations between diameter and signal intensity, indicating that the measurement is affected by intensity changes, whereas model-based methods show weak or no correlation.

375 This finding has important implications for longitudinal studies and any experiment where
 376 angiogram intensity may vary between or within imaging sessions. In longitudinal mouse
 377 imaging, intensity losses are common due to cataract formation, corneal drying, or changes
 378 in anaesthesia depth. In clinical settings, factors such as media opacities, pupil dilation, and
 379 signal attenuation can similarly affect angiogram intensity. If a binary mask based method is
 380 used under such conditions, apparent changes in vessel diameter may be artefactual, driven by
 381 intensity fluctuations rather than true vascular changes. Model-based methods are robust to
 382 these intensity variations because they fit the shape of the intensity profile rather than relying on
 383 absolute intensity values to define vessel boundaries.

384 4.5. Literature Comparison

385 Hormel et al. [12] showed that MIP produces en face OCTA images with higher SNR and contrast
 386 compared to MeIP, and concluded that MIP should be preferred for OCTA visualisation. Our
 387 results are consistent with this for image quality purposes, but reveal a different picture when it
 388 comes to quantitative vessel diameter measurement. We found that MeIP consistently yielded
 389 more accurate and precise diameter values than MIP across both model-based and binary mask
 390 based approaches. This suggests that the projection method best suited for visualisation is not
 391 necessarily the best suited for quantification, and that the choice of projection should depend on
 392 the intended analysis.

393 The challenge of standardising OCTA image analysis has been highlighted by Sampson et al. [13]
 394 and Untracht et al. [11], both of whom emphasised the need for transparent, reproducible analysis
 395 pipelines. Our findings add to this discussion by showing that the choice of diameter measurement
 396 method introduces a source of variability that is at least as significant as the choice of binarisation
 397 threshold. The model-based approach presented here offers a path toward more standardised
 398 vessel diameter quantification, as it reduces the dependence on operator-selected thresholds and
 399 produces measurements that are more robust across imaging conditions.

400 The RBC orientation artefact and its effect on OCTA vessel appearance has been described by
 401 Bernucci et al. [15], who used Intralipid contrast to demonstrate that the hourglass pattern in
 402 vessel cross-sections is caused by the directional scattering properties of RBCs rather than by

403 true variations in flow. Our work extends this observation by quantifying the degree to which
404 this artefact affects diameter measurements and by proposing model-based fitting as a practical
405 correction strategy. While contrast enhancement with Intralipid is effective for revealing the
406 true vessel diameter, it requires a contrast injection and is not yet applicable in clinical settings.
407 Using a calibrated model-based approach provides a way to partially recover the underestimated
408 diameter from native OCTA data without the need for a contrast agent.

409 Recent work by Son et al. [18] using functional OCT to image flicker-evoked vasodilation
410 in the mouse retina showed that vasodilation is anisotropic, with larger changes in the axial
411 than the lateral direction. Our flicker stimulation results complement this by demonstrating that
412 the choice of diameter measurement method can determine whether the vascular response is
413 correctly detected at all. The fact that binary mask based methods reported vasoconstriction
414 where model-based methods correctly detected vasodilation underscores the importance of
415 measurement methodology when studying neurovascular coupling and other functional vascular
416 responses.

417 4.6. *Future Directions*

418 A future direction of this work is to move toward fully automated segmentation and calculation of
419 vessel diameters from en face angiogram maps. While ROIs were manually selected in this work,
420 it is also possible to automatically segment angiographic data using traditional skeletonisation
421 methods, which are commonly used for automated OCTA analysis. From a skeletonised image,
422 vessel angle can be computed for further extraction of vascular cross-sections perpendicular to
423 the vessel axis. These cross-sections could then be processed using the model-based methods
424 described here to enable fully automated analysis of 3D OCTA data sets, rather than the semi-
425 automated approach shown here for 2D OCTA data. This type of model-based approach is
426 particularly important for comparing 3D volumes acquired at different time points in the mouse
427 eye, as there are often intensity losses caused by drying of the eye, cataract formation, etc. As we
428 have shown in this work, even low levels of intensity loss occurring over a short time window
429 can significantly affect the measured vessel diameter when using a binarisation approach, but not
430 when using the models presented here.

431 Additionally, this work focused on the larger vessels of the superficial vascular plexus.
432 Extending the model-based approach to the intermediate and deep capillary plexuses, where
433 vessels are smaller and signal levels are lower, would be valuable for studying microvascular
434 changes in disease. For very small vessels approaching the lateral resolution limit of the imaging
435 system, the measured intensity profile is a convolution of the true vessel profile with the point
436 spread function (PSF) of the incident beam on the retina. Since the lateral resolution of OCT is
437 typically worse than the axial resolution — being determined by the beam optics rather than the
438 source bandwidth — this convolution becomes increasingly significant for smaller vessels. If the
439 PSF were known, a deconvolution approach could be used to recover the true vessel profile before
440 model fitting, potentially improving accuracy for capillary-scale vessels. However, characterising
441 the retinal PSF in vivo remains challenging, as it depends on the ocular optics of each individual
442 eye and may vary with retinal depth and eccentricity.

443 4.7. *Computation Time*

444 It is clear from the results that model-based approaches outperform traditional diameter measure-
445 ments based on a binarised image. Nevertheless, the improved linearity and repeatability come at
446 the cost of computation time. Fitting a model to the raw angiogram data is significantly more
447 computationally expensive and may not always be the most appropriate choice if results of the
448 processing are time sensitive. Nevertheless, these approaches are highly parallelisable given
449 the independent nature of the measurements, so it is expected that computation time could be
450 significantly accelerated with appropriate hardware.

451 4.8. Thresholding

452 The results presented here used a basic, global threshold for the binarisation of the angiogram
 453 data. There are numerous other approaches for binarising the angiogram, which will most likely
 454 each behave differently. For example, angiogram signals can be enhanced with vesselness filters
 455 and dynamic thresholds can be used to binarise images with varying SNR. When applying a
 456 vesselness filter, it is possible to enhance the shape of the angiogram and omit noise, but it is
 457 also possible for such enhancement to overcompensate the signal, leading to larger than expected
 458 vessel appearance. Dynamic thresholding can also yield a more uniform angiogram appearance,
 459 but similarly risks over- or undercompensating vessels given that threshold level directly affects
 460 the measured diameter. Further analysis of these other methods would be required to determine
 461 their efficacy in returning accurate and precise diameter values.

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